Unit I: Diversity of Living World

Chapter 2

Plant Kingdom

Learning Objectives

The learner will be able to,

- *Outline the classification of plants*
- *Illustrate the life cycles in plants*
- *Recognize the general characteristic features and reproduction of Algae*
- *Describe the structure, reproduction of Chara*
- *Recognize the general characteristic features of Bryophytes*
- *Describe the structure, reproduction of Marchantia*
- *Recognize the general characteristic features of Pteridophytes*
- *Describe the structure, reproduction of Selaginella*
- *Describe the general characteristic features of Gymnosperms*
- *Explain the structure, reproduction of Cycas*
- *Recognize the salient features of Angiosperms*

Chapter Outline

2.1 Classification of Plants

2.2 Life Cycle patterns in Plants

2.3 Algae

2.4 Bryophytes

2.5 Pteridophytes

2.6 Gymnosperms

2.7 Angiosperms

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Traditionally organisms existing on the earth were classified into plants and animals based on nutrition, locomotion and presence or absence of cell wall. Bacteria, Fungi, Algae, Bryophytes, Pteridophytes, Gymnosperms and Angiosperms were included under plant group. Recently, with the aid of molecular characteristics. Bacteria and Fungi were segregated and placed under separate kingdoms. Botany is one of the oldest science in the world because its origin was from time immemorial as early men explored and identified plants for the needs of food, clothing, medicine, shelter etc., Plants are unique living entities as they are endowed with the power to harvest the light energy from the sun and to convert it to chemical energy in the form of food through the astounding reaction, **photosynthesis**. They not only supply nutrients to all living things on earth but sequester carbon-di-oxide during photosynthesis, thus minimizing the effect of one of the major green house gases that increase the global temperature. Plants are diverse in nature, ranging from microscopic algae to macroscopic highly developed angiosperms. There are mysteries and wonders in the plant world in terms of size, shape, habit, habitat, reproduction etc., Although plants are all made up of cells there exists high diversity in form and structure (Table 2.1).

2.1 Classification of Plants

Classification widely accepted for plants now include Embryophyta which is divided into Bryophyta and Tracheophyta. The latter is further divided into Pteridophyta and Spermatophyta (Gymnospermae and Angiospermae). An outline Classification of Plant Kingdom is given in Figure 2.1

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